SentinaAI

The **AI Intelligence** Layer Behind **Modern Convention Centers**

@rumaisa zia

The Problem

High-volume lot data
from CCTV, HVAC, lighting, access control & sensors

Cloud-only systems
cause latency & slow incident response

Disconnected tools
make operations reactive, not coordinated

System Architecture

Edge Devices

- Access Point
- CCTV Cameras
- Occupancy Sensor
- HVAC

Edge Node

- MQTT Broker
- Local Inference
- Rule Engine
- Local Cache (Offline Mode)

Cloud Layer

- Analytics Engine
- Al/ML Models
- Orchestration & APIs
- Authentication/ RBAC

Client Layer

- Security Dashboard
- Operations Dashboard
- Sustainability Dashboard
- Exhibitor Portal

Security

- **Multi-source threat** correlation
- **2.5s real-time alert** polling
- **JWT-enforced RBAC** audit trail
- **Zone-level security** hotspot analytics

Sustainability

- **Energy Usage** & HVAC Efficiency
- **Carbon Emission** Estimates
- **Environment Health** Score
- **Hall-level Energy** Breakdown

Machine learning algorithms:
- RandomForest Regressor
- RandomForestClassifier

Why Hybrid Edge-Cloud?

- **Low-latency** real-time inference
- **Centralized analytics** & model lifecycle
- **Resilient operations** even offline

Operations

- **Live Occupancy** & Comfort Index
- **Crowd Flow** & Congestion
- **Busiest Halls &** Al Live Status
- **Predictive Occupancy**

Machine learning algorithms:
- RandomForest Regressor
- RandomForestClassifier

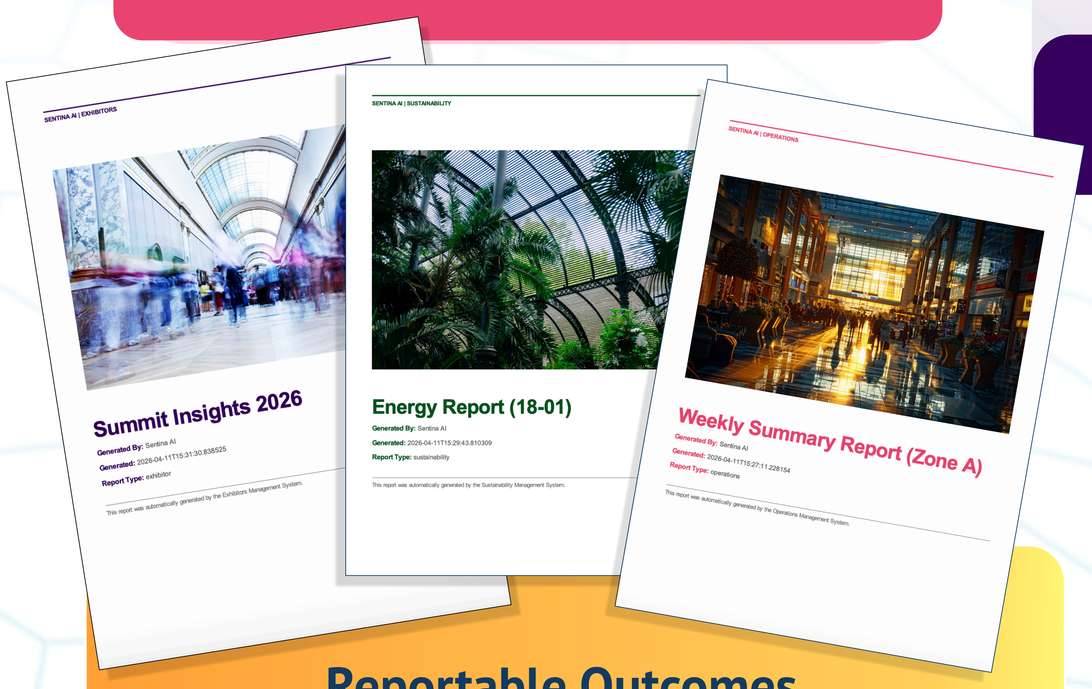
Exhibitors

- **Assigned Booth** & Al Confidence
- **Engagement Trends & Hall Rankings**
- **Competition Density**
- **Catchment Heatmaps**

Machine learning algorithms:
- Spatial-Temporal GCN (per timestep) + LSTM (across time)

Digital Twin

- **Real-time IoT** sensor fusion
- **Sub-second telemetry** updates
- **Edge-synced 3D** twin



Reportable Outcomes

- **Draft, preview, and generate** workflows
- **Role-based** report templates
- **PDF and XLSX export**

Guided Assistant

- **Structured workflows,** not open-ended chat
- **Role-aware assistant** with structured analysis options (rule based)

Navigation

- **Live occupancy** heatmap rendering
- **Crowd-aware** dynamic routing

Testing Scope

- Python Unittest Scripts
- Paho-mqtt with Py Unittest

Tech Stack

Node JS · Express JS · Postgress · MQTT Broker · Tensorflow · React · Open CV · PyTorch · Ultralytics